

Conflict, not confidence: which value $\rightarrow$ RT mapping best describes probability discounting, and does RT help?

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The valuation model, choice rule, comparison framework and report were developed jointly; the three authors contributed equally. Requested assessment: full grading.		

Abstract We model the choices and the reaction times of 49 participants in a probability discounting task with reward and loss conditions. One valuation model is held fixed: a Rachlin hyperboloid in the odds against reward, $V_{unc} = r_{unc}/(1 + h\Theta)$, with separate rates for gains and losses, coupled to choice by a logistic rule on the normalised value difference ΔV . Three RT models sit on top of it and differ only in how ΔV enters the location of a shifted log-normal: linear conflict ($|\Delta V|$), decision uncertainty ($|2P - 1|$) and quadratic conflict (ΔV^2). Estimation is per participant by MAP on run A, with run B held out. On the quantity that separates the mappings, the run B RT predictive log-likelihood, they are statistically indistinguishable (all pairwise $p > 0.15$), and the in-sample ranking that favours decision uncertainty fails to generalise. Against a choice-only baseline fitted to the identical model, adding RT never improves held-out choice prediction: the choice predictive log-likelihood falls from -0.630 to -0.640 , -0.643 and -0.640 nats per trial. The gain is elsewhere. In simulation, quadratic conflict halves the recovery error of both discount rates, and on real data it alone preserves test–retest reliability, holding $\log h_{gain}$ at 0.447 against a baseline of 0.444 and lifting $\log h_{loss}$ from 0.124 to 0.304 , where the other two mappings roughly halve the reliability of $\log h_{gain}$. We recommend quadratic conflict: modelling RT is worth doing to estimate the discounting constructs, but not to predict choices. Non-decision time is not identifiable.

1 INTRODUCTION

Every trial of a value-based choice task yields two observations: which option was taken, and how long the participant took to take it. Discounting research almost always uses only the first, largely because discount rates are traditionally estimated from titrated indifference points, which summarise a block of choices and discard their timing, and because the discounting models themselves are static value functions with no dimension along which a response time could be predicted (Green & Myerson, 2004). Both modalities are nevertheless driven by the same latent quantity: if the subjective value V drives the choice, it should also govern how long the comparison takes, making RT a second and essentially free measurement of the same construct. If RTs carry information about h and β , fitting both modalities should sharpen those estimates; if they do not, the joint fit pays for the RT term and gets nothing back.

We therefore hold valuation and choice entirely fixed, using a Rachlin hyperboloid in odds against (Rachlin et al., 1991) with a logistic choice rule, and vary only the value $\rightarrow$ RT mapping. Linear conflict (M1) treats RT as tracking the distance between the options in value space, the first-order behaviour of an accumulator with drift proportional to ΔV (Ratcliff, 1978; Ratcliff & McKoon, 2008). Decision uncertainty (M2) treats it as tracking subjective indecision, so the effect saturates once a participant is confident. Quadratic conflict (M3) concentrates the slowing near indifference. The three span the plausible curvatures, and the report asks in turn: which mapping predicts RT best out of sample, whether its parameters are identifiable and reliable, and whether modelling RT improves our account of the choices relative to the same model fitted to choices alone.

2 METHODS

2.1 Data

The probability discounting data set contains 49 participants, each with an inference run (A) and an evaluation run (B) of 200 trials. On each trial a certain reward r_{cert} is offered against an uncertain reward r_{unc} paid with

probability $p \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$, which we summarise by the odds against, $\Theta = (1 - p)/p$. A reward condition (both magnitudes positive) and a loss condition (both negative, verified in the raw files, so no sign correction is applied anywhere) are fully crossed with the design at 100 trials per condition per run.

We drop 68 non-responses (0.35% of 19,600 trials) and 105 trials faster than 250 ms (0.54%), a fixed global floor for anticipatory responses, leaving 19,427. Nothing is trimmed from the upper tail: the extreme RTs are 200 ms and 9996 ms with no pile-up at either edge, so there is no sign of censoring. Four participants make more than 10% of their responses below 400 ms; we flag them and keep them. Participants choose the uncertain option on 46.4% of reward trials and 43.2% of loss trials (between-participant SDs 0.14 and 0.10), respond more slowly under losses (median 2.09 s) than gains (1.81 s), and differ enormously in overall speed (participant median RTs span 0.40–3.50 s). Raw RT is strongly right-skewed (1.55) and near-symmetric on the log scale (-0.51), which is what recommends a log-normal observation model.

Before fitting we check the valuation model model-free. A logistic of $\log(r_{unc}/r_{cert})$ fitted separately at each p and condition gives the indifference ratio at that Θ , and a hyperboloid predicts $r_{unc}/r_{cert} = 1 + h\Theta$, linear in Θ . Figure 1 shows that it is, with slopes $h \approx 3.24$ for gains and $h \approx 0.35$ for losses: the order-of-magnitude gap is the model-free signature of the reflection effect (Kahneman & Tversky, 1979; Green & Myerson, 2004).

One property of the two runs matters throughout, so we establish it here. Run B is not a replication of run A. Its offers move to larger magnitude ratios (mean log ratio 1.73 against 1.32), and 23% of run B trials fall outside the range that participant saw in run A. Behaviour moves too. Participants choose the uncertain option more often (0.394 to 0.503, $p = 0.001$); most of that is carried by the changed offers, but a residual $+0.038$ survives matching participant, condition, p and ratio octile ($p = 0.0015$). They also speed up sharply, median RT falling from 2.19 s to 1.77 s with 45 of 49 faster ($p < 10^{-11}$). Section 2.3 fixes τ and a from run A, so nothing absorbs a session-wide speed-up.

2.2 Models

Valuation and choice (held fixed). The certain option needs no discounting, so $V_{cert} = r_{cert}$, while the uncertain option is discounted by the odds against it (Rachlin et al., 1991):

$$V_{unc} = \frac{r_{unc}}{1 + h\Theta}, \quad h \geq 0, \quad (1)$$

with separate rates h_{gain} and h_{loss} per participant. Because loss magnitudes are stored negative, this single equation produces both risk attitudes at no extra cost in parameters: V_{unc} shrinks towards zero, which is worse than a sure gain (risk aversion) but better than a sure loss (risk seeking), so the reflection effect emerges for free. Magnitudes span five orders of magnitude, so we normalise, $\Delta V = (V_{unc} - V_{cert}) / [(|r_{cert}| + |r_{unc}|)/2]$, which lets one sensitivity serve small and large trials at the price of removing absolute magnitude effects. Choice is logistic, $P_t \equiv P(\text{uncertain}) = \sigma(\beta \Delta V_t + \beta_0)$, where β_0 absorbs a value-independent bias, giving the Bernoulli trial likelihood $\log p(y_t | \theta) = y_t \log P_t + (1 - y_t) \log(1 - P_t)$ with $y_t = 1$ when the uncertain option was chosen.

RT distribution (held fixed). RTs follow a shifted log-normal, $RT = \tau + \exp(\mu + \sigma\varepsilon)$ with $\varepsilon \sim \mathcal{N}(0, 1)$, so for $rt > \tau$

$$\log p(rt | \mu, \sigma, \tau) = -\log(rt - \tau) - \log \sigma - \frac{1}{2} \log 2\pi - \frac{1}{2} \left[\frac{\log(rt - \tau) - \mu}{\sigma} \right]^2. \quad (2)$$

Holding this distribution identical across the three RT models is what makes the comparison one of mappings rather than of distributional shape. The full trial likelihood is $p(\text{choice}, RT | \theta) = p(y_t | \theta) p(rt_t | \theta)$: the modalities are conditionally independent given the shared latent ΔV , and that shared valuation is what makes the model multimodal. The models differ only in the location,

$$\text{M1: } \mu_t = a - b |\Delta V_t| \quad \text{M2: } \mu_t = a - b |2P_t - 1| \quad \text{M3: } \mu_t = a - b \Delta V_t^2 \quad (3)$$

Psychological hypotheses. M1 says what slows a participant is the distance between the options in value space, every extra unit of advantage buying the same amount of speed, which is the first-order behaviour of an accumulator with drift proportional to ΔV (Ratcliff & McKoon, 2008; Krajovich et al., 2010). M2 says the driver is subjective indecision instead. Its predictor is 0 under complete indecision and 1 under certainty, so once

$P \approx 0.95$ further advantage buys no more speed: M1 has RT tracking value, M2 has it tracking confidence. M3 says deliberation is costly only very near indifference, making the marginal speed-up grow rather than shrink with $|\Delta V|$. The psychological claim in each case is the curvature.

Choice-only baseline. The baseline deletes the RT term from the likelihood and fits the identical valuation model and choice rule to the choices alone. It is not a fourth model but the same model fitted to one modality, and it is the reference against which the RT modality is valued.

2.3 Inference

We estimate parameters per participant by maximum a posteriori on run A, leaving run B untouched at this stage:

$$\log p(\theta \mid \text{data}) \propto \underbrace{\sum_t [y_t \log P_t + (1 - y_t) \log(1 - P_t)]}_{\text{choice}} + \underbrace{\sum_t \log p(rt_t \mid \mu_t, \sigma, \tau)}_{\text{RT; omitted in the baseline}} + \log p(\theta). \quad (4)$$

Four choice parameters are fitted unconstrained on transformed scales (`log_h_gain`, `log_h_loss`, `log_beta`, `beta0`), and the RT models add `a`, `b`, `log_sigma` and `tau_raw`, where $\tau = \sigma(\text{tau_raw}) \cdot 0.99 \min(RT)$ enforces $0 < \tau < \min(RT)$. The priors are weakly informative Gaussians on the fitted scale (Table 1), wide enough that the data dominate wherever the likelihood has curvature. The prior on b is centred at zero and symmetric, so a positive difficulty effect is never assumed, and M3's is widened to $\mathcal{N}(0, 3^2)$ because its predictor is on a smaller scale (median 0.08 against 0.28). That equalises regularisation in terms of the effect on μ rather than the raw coefficient; without it M3 would be penalised hardest and the comparison would be unfair.

Optimisation uses L-BFGS-B with box constraints and 10 restarts per participant, one from the model-free estimates above and the rest drawn from the priors; seeds derive deterministically from (model, run, participant), so the pipeline reproduces exactly. The optimiser reports convergence for every fit except M2 on run A (96%). Two diagnostics guard the estimates. Refitting by MLE with the priors off returns essentially identical parameters for well-identified participants ($r \geq 0.995$, median deviation ≤ 0.13), so the priors are not doing the work. A curvature check, the log-likelihood lost when each parameter is perturbed by ± 0.5 , flags 19 of 49 participants as poorly identified in at least one parameter (14 in $\log h_{\text{gain}}$, 14 in $\log h_{\text{loss}}$, three also fast responders); we keep all 19 and track them separately.

2.4 Evaluation Plan

Recovery. For each RT model we draw 200 parameter sets covering the run A estimates, simulate choices and RTs over a real participant's design, and refit twice, with the full choice + RT likelihood and with the choice-only likelihood. Correlation and RMSE between true and estimated values quantify recovery; the correlation structure of the errors diagnoses identifiability. *Reliability.* We estimate the same model independently on runs A and B and take test-retest reliability as the across-participant correlation of the estimates (Pearson, Spearman, ICC(2,1) (Shrout & Fleiss, 1979)) with 2000-sample bootstrap CIs. CIs for differences resample participants once and apply that resample to both models, preserving the pairing. *Prediction.* Run A parameters are applied unchanged to run B, with nothing refitted. For choice we report predictive log-likelihood per trial, accuracy, balanced accuracy and AUC; for RT, predictive log-likelihood per trial and RMSE on $\log(RT - \tau)$; each for all trials and separately by condition. *Which quantity compares what.* A joint likelihood and a choice-only likelihood describe different data, so their totals are not comparable. We compare the three RT models with each other on the RT predictive log-likelihood only, and the RT-informed fits with the choice-only baseline on choice quantities only. Group comparisons use paired Wilcoxon signed-rank tests.

3 RESULTS

The fits agree with the model-free analysis. Median run A baseline estimates are $h_{\text{gain}} = 2.65$ (IQR 1.00–7.99), $h_{\text{loss}} = 0.82$ (0.21–1.98), $\beta = 4.53$ and $\beta_0 = -0.24$, with $h_{\text{gain}} > h_{\text{loss}}$ in 32 of 49 participants, so the reflection effect is recovered by the fit as well as by the raw indifference points. The wide IQRs are a first sign of how heterogeneous the sample is.

3.1 Parameter Recovery and Identifiability

Every choice parameter recovers well under every RT model (Fig. 2, Table 2). Correlations run 0.90–0.98 for $\log h_{gain}$, 0.94–0.98 for $\log h_{loss}$, and at least 0.95 for $\log \beta$ and β_0 ; a , b and $\log \sigma$ also recover well (0.83–0.98). Non-decision time is the exception. τ is not identifiable, with $r = 0.19$ (M1), 0.21 (M2) and 0.29 (M3) and an upward bias of about +0.05 s. The recovery errors show why: $\text{corr}(\text{err}_\tau, \text{err}_a) = -0.64$ under M1 and M3. Both parameters shift the predicted RT distribution, and with a unimodal log-normal and roughly 200 trials the data cannot separate them, so an over-estimated shift is paid for by an under-estimated intercept. Nothing else trades off appreciably (off-diagonal $|r| \leq 0.31$, Fig. 3).

Adding RT improves recovery of the choice parameters in every case, but by amounts that differ enormously between mappings. As a ratio of RMSE (joint over choice-only, so below 1 means RT helps), M3 gives 0.47 for $\log h_{gain}$ and 0.49 for $\log h_{loss}$, roughly halving the error, against 0.86 and 0.72 for M1 and 0.98 and 0.88 for M2. Paired Wilcoxon tests on absolute errors confirm the improvement under M1 and M3 ($p < 10^{-4}$) and only weakly under M2 ($p = 0.03$ and 0.16). In simulation RT does change identifiability, most under M3.

3.2 Reliability

Test–retest reliabilities are moderate at best (Fig. 4, Table 3). Under the choice-only baseline $\log \beta$ is the most stable construct ($r = 0.575$, CI [0.36, 0.76], ICC 0.52), followed by $\log h_{gain}$ (0.444, [0.15, 0.68]) and β_0 (0.361), while $\log h_{loss}$ is essentially unstable (0.124, [−0.17, 0.39]). Values in this range are normal for cognitive-task parameters, whose between-subject reliability sits far below that of questionnaires (Hedge et al., 2018; Enkavi et al., 2019). Two features of the data push them down further: 19 participants are poorly identified, and, as Section 2.1 showed, behaviour itself shifted between runs. Our reliabilities are therefore a lower bound on the stability of the constructs, since some of the disagreement between runs is real change rather than estimation noise.

The effect of adding RT is the central and least expected result. Under M1 and M2 reliability falls sharply: $\log h_{gain}$ from 0.444 to 0.212 and 0.187, β_0 from 0.361 to 0.092 and 0.164. Bootstrap CIs on the paired difference are [−0.52, +0.06] for M1 and [−0.53, −0.01] for M2, so for M2 the loss is reliable. Under M3 nothing falls. $\log h_{gain}$ is unchanged (0.447 against 0.444, $\Delta r = +0.003$, [−0.21, +0.23]) and $\log h_{loss}$ more than doubles (0.304 against 0.124, $\Delta r = +0.180$, [−0.04, +0.40]). The RT parameters are themselves among the most reliable quantities anywhere in the analysis (a : 0.62–0.73; $\log \sigma$: 0.67; τ : 0.39–0.41), whereas the difficulty slope b is not reliable at all (0.09–0.20): participants have a stable overall speed and spread, but no stable sensitivity of RT to difficulty. Restricting to the 30 well-identified participants recovers part of the loss under M1 ($\log h_{gain}$: 0.21 to 0.42), so poor identification amplifies the effect without creating it.

3.3 Predictive Performance (Choice and RT, by Condition)

RT. On the run B RT predictive log-likelihood the three models are indistinguishable (Fig. 5, Table 4): M2 −1.524, M3 −1.544, M1 −1.548 nats per trial, all pairwise $p > 0.15$ (M1 vs M2 $p = 0.84$, M2 vs M3 $p = 0.41$, M1 vs M3 $p = 0.16$). The ordering is not even stable across conditions, with M2 nominally best on reward trials and M1 on loss trials. In sample on run A the picture looked quite different: M2 beat M1 ($p = 0.005$) and M3 ($p = 0.014$) and was best for 29 of 49 participants. That advantage disappears entirely out of sample. Note also that every model loses about 0.24 nats per trial moving from run A to run B, far more than any difference between models. Section 2.1 explains it: participants speed up by a median of 0.40 s in run B, τ and a are fixed from run A, and no component of the model can absorb a session-wide change in speed.

Choice. All models predict run B choices well above chance ($\log 0.5 = -0.693$). The baseline reaches −0.630 nats per trial, 70.4% accuracy, 69.9% balanced accuracy and AUC 0.756, and is well calibrated, though performance varies widely across participants (AUC 10th to 90th percentile 0.56–0.91, 7 below 0.60) and is slightly better for loss than reward trials (−0.623 against −0.637). Every RT-informed fit is slightly worse on choice than the baseline, in every condition.

3.4 Value of the RT Modality

(a) *Out-of-sample choice prediction: RT does not help.* Measured on the choice predictive log-likelihood, all three RT models lose to the baseline: $\Delta = -0.0099$ (M1, $p = 0.12$), -0.0127 (M2, $p = 0.053$) and -0.0100

(M3, $p = 0.11$) nats per trial, with only 18 to 20 of 49 participants improved (Fig. 6). Balanced accuracy (0.699 to 0.689–0.692) and AUC agree, in both conditions separately. No single loss reaches $p < 0.05$, M2 coming closest, but the direction is unanimous across models, metrics and conditions.

(b) Identifiability and reliability: RT helps, but only under M3. Simulation says RT should help a great deal (§3.1), yet on real data M1 and M2 make the discounting parameters markedly less reliable (§3.2). Effect size reconciles the two. The simulation draws b from ranges implying a strong difficulty effect, whereas the effect in the data is close to null: the within-participant slope of $\log RT$ on $|\Delta V|$ is -0.027 and the raw pooled slope $+0.063$, so even the sign depends on how participant-level RT offsets are handled, and median fitted b is only weakly positive ($+0.11$, $+0.30$, $+0.15$) and negative in 20, 11 and 16 of 49 participants. In that regime an RT term mostly injects variance into the shared parameters. M3 escapes because ΔV^2 concentrates nearly all its leverage on near-indifference trials, exactly the trials whose ΔV the choices already constrain tightly. M1 and M2 spread leverage over easy trials, where ΔV is poorly constrained because any large difference predicts the same response, and so drag h around to fit RT variance that carries no choice information.

Net. Modelling RT gains a better-identified and, under M3, more stable estimate of the discounting constructs, plus an account of a second behavioural modality. It costs roughly 0.01 nats per trial of choice prediction, indistinguishable from zero under M1 and M3 and borderline under M2, and under M1 and M2 about half the test–retest reliability of $\log h_{gain}$.

4 DISCUSSION

We recommend M3, quadratic conflict. The RT data by themselves cannot pick a mapping: the three sit within 0.02 nats per trial of one another out of sample with all pairwise $p > 0.15$, the ordering flips between conditions, and the in-sample winner does not generalise. The tie has to be broken on the rest of the evidence, and there M3 is unambiguous. It recovers both discount rates best ($r = 0.977$ and 0.978), halves their recovery error relative to a choice-only fit, and is the only mapping that leaves real-data reliability intact, holding $\log h_{gain}$ and doubling $\log h_{loss}$, at a choice-prediction cost indistinguishable from zero. Psychologically that favours deliberation being expensive only near indifference, rather than scaling with value distance or tracking confidence.

Was modelling RT worthwhile? Not for predicting choices, where no mapping improves on the baseline and M2 comes close to significantly worsening it ($p = 0.053$). For estimating the constructs, yes, but only under M3. If the goal is choice prediction, drop RT; if it is a stable, well-identified discount rate, which is usually why these models are fitted, M3 earns its place.

What we could not establish, and limitations. We cannot claim any mapping is the true one. The empirical difficulty effect is weak and its sign unstable, so the comparison had little power, and the simulation therefore describes what RT could contribute where difficulty genuinely modulates RT rather than what it contributes here. Four limitations bound the conclusions. First, the RT model is descriptive rather than mechanistic: a shifted log-normal with a value-driven location cannot separate drift from boundary, so a speed–accuracy shift and a change in valuation are confounded in b . That is the likeliest reason RT can actively damage h , since a genuine accumulator would let the boundary absorb the overall speed level. Second, the two runs are not exchangeable: offers move to larger ratios, 23% of run B trials fall outside the participant’s own run A range, and responses speed up by a median of 0.40 s, so out-of-sample performance mixes model error with extrapolation and session change, while reliability is attenuated by real drift. Third, τ is unidentified and anchored to a single extreme observation per participant and run, and run B RTs below run A’s τ (0.25–0.28% of trials) are floored at $LL = -20$, contributing about 0.05 nats per trial, the same order as the gaps between RT models, so the nominal $M2 > M3 > M1$ ordering is not robust. M2 is also the least numerically stable model, with the flattest optimum (96% convergence against 100% elsewhere) and estimates that shift slightly across SciPy versions. Fourth, estimation is per participant with no pooling, one β is shared across conditions, ΔV is normalised so magnitude effects are removed by construction, and parameters are held constant within a run.

Next steps. First, fit a genuine sequential-sampling model, DDM or LBA, in which ΔV drives drift and a separate boundary carries speed–accuracy (Rodriguez et al., 2014; Dai & Bussemeyer, 2014). That directly tests whether RT harms the discount rates because the signal is absent or because our functional form is wrong, and a free boundary would also absorb the run B speed-up that costs every model 0.24 nats per trial. Second, estimate hierarchically with partial pooling on $(\log h_{gain}, \log h_{loss}, \log \beta, b)$, which should stabilise the 19 poorly-identified participants and sharpen the reliability comparison.

FIGURES

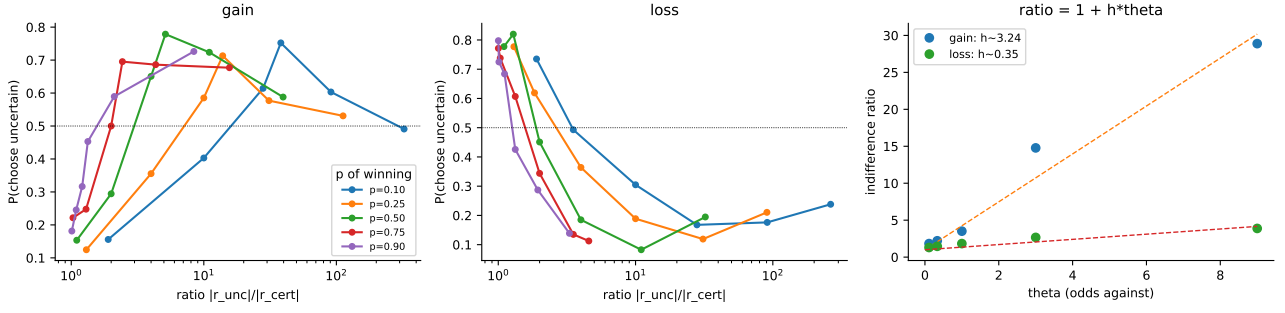


Figure 1: **Model-free data summary.** Left and middle: proportion of uncertain choices against the magnitude ratio $|r_{unc}|/|r_{cert}|$, binned into sextiles, separately for each outcome probability p , in the reward (left) and loss (middle) conditions. Right: indifference ratios estimated at each p against the odds against Θ . The hyperboloid predicts $r_{unc}/r_{cert} = 1 + h\Theta$ (dashed lines), giving model-free $h \approx 3.24$ for gains and $h \approx 0.35$ for losses.

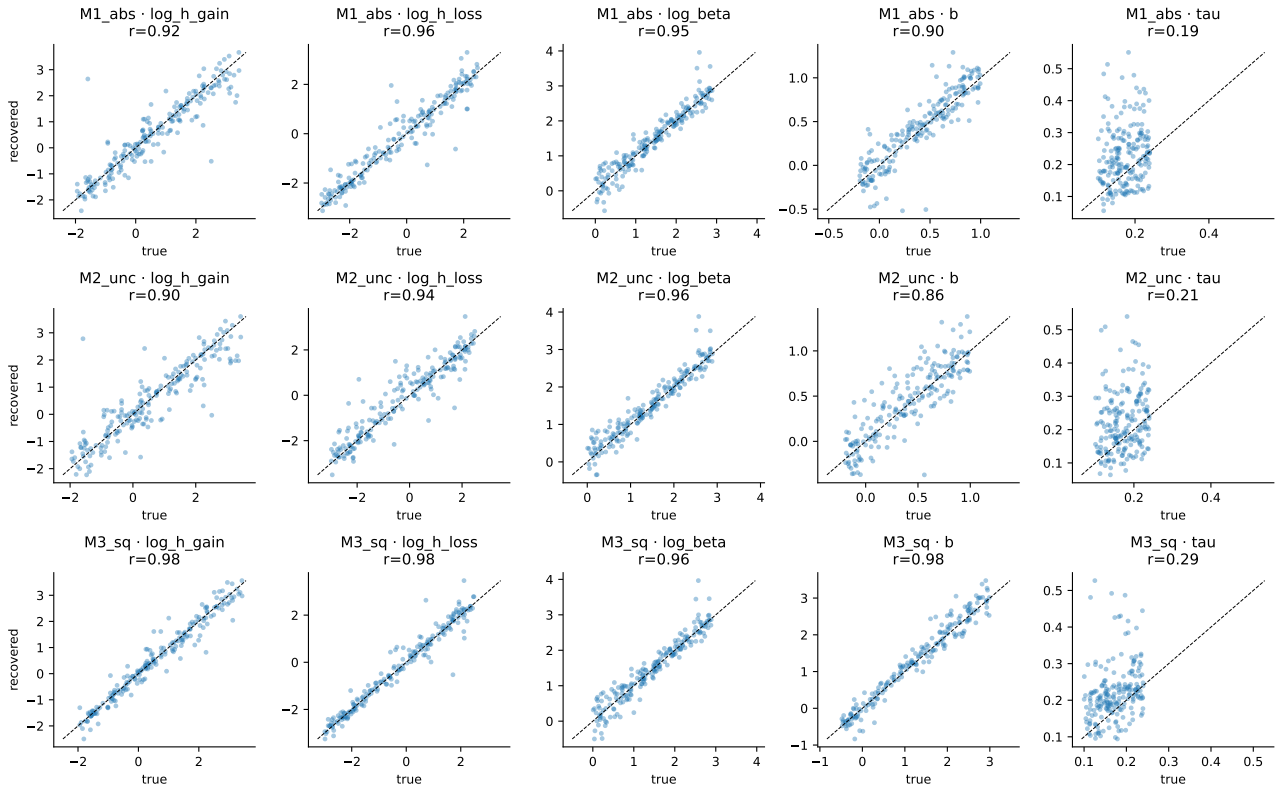


Figure 2: **Parameter recovery.** True against recovered values over 200 simulated participants per model (rows: M1, M2, M3) for the two discount rates, choice sensitivity, the difficulty slope b and non-decision time τ . Dashed line is identity. All choice parameters recover well; τ does not ($r = 0.19$ – 0.29).

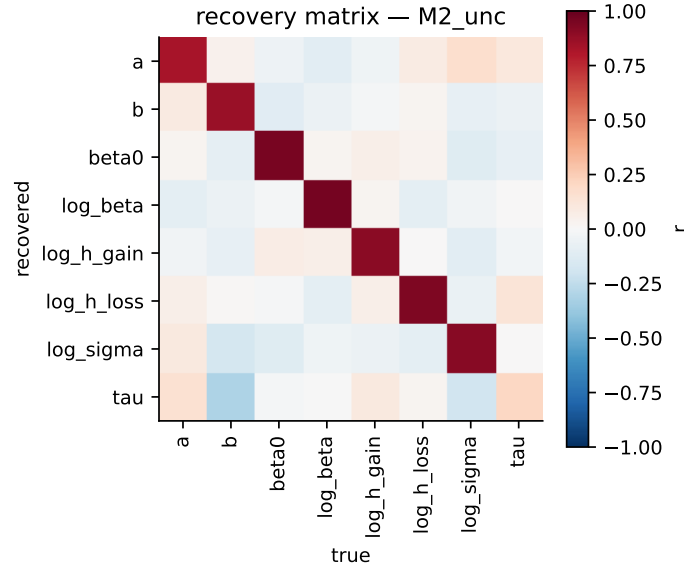


Figure 3: **Recovery matrix (M2).** Correlation between each true parameter (columns) and each recovered parameter (rows). Off-diagonal structure indicates trade-offs; the only appreciable one is τ against b and a .

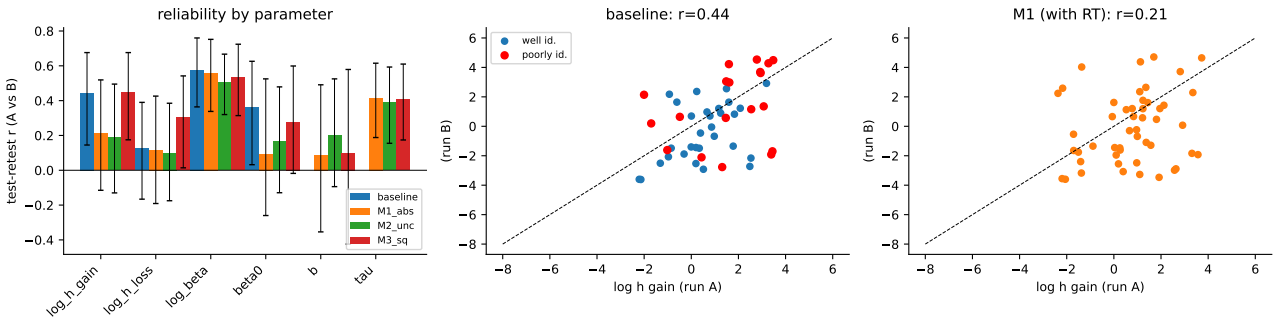


Figure 4: **Test-retest reliability (run A against run B).** Left: reliability per parameter and model with bootstrap 95% CIs. Middle and right: run A against run B estimates of $\log h_{gain}$ for the choice-only baseline (red marks participants flagged as poorly identified by the curvature diagnostic) and for M1, illustrating the loss of stability when RT is added under the linear mapping.

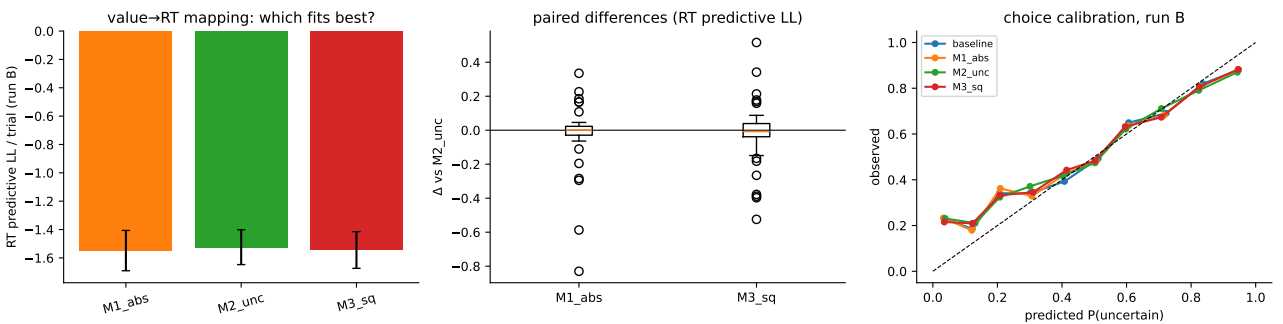


Figure 5: **Out-of-sample prediction (run B).** Left: RT predictive log-likelihood per trial by model (mean $\pm$ SEM across participants), the common quantity on which the RT models are compared. Middle: paired per-participant differences against the nominally best model. Right: calibration of predicted against observed choice probability, decile-binned.

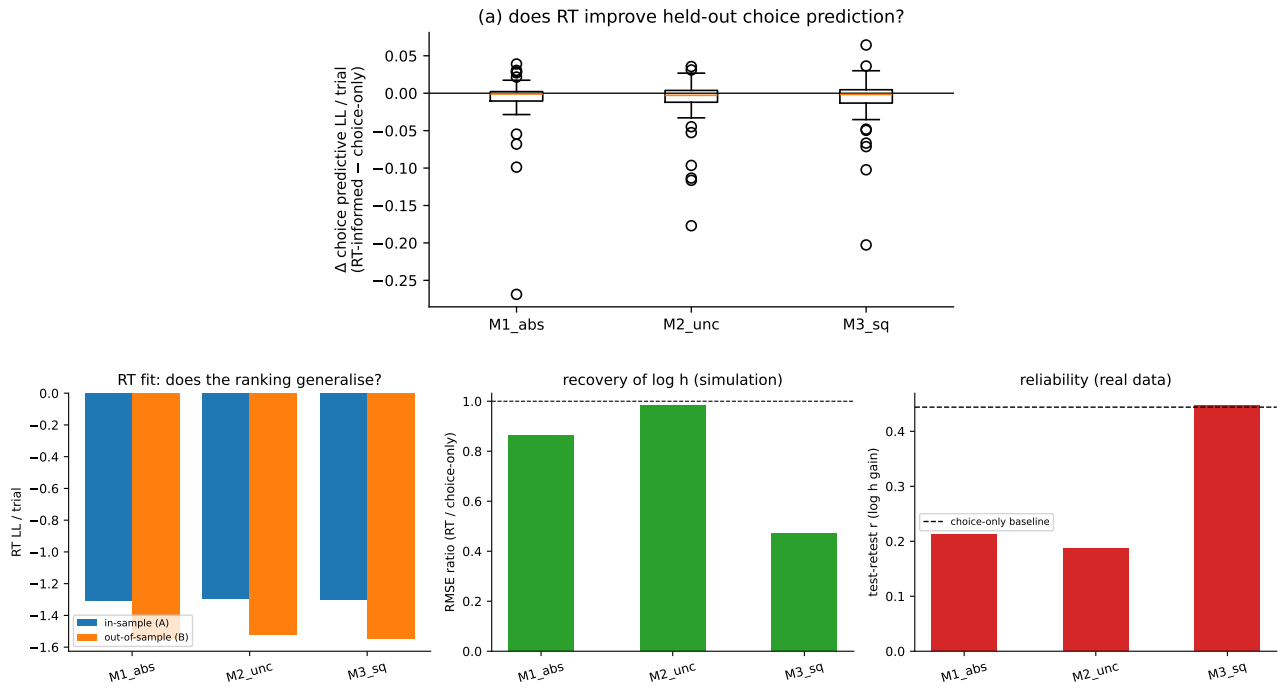


Figure 6: **Value of the RT modality.** Top: per-participant difference in run B choice predictive log-likelihood between each RT-informed fit and the choice-only baseline; values below zero mean RT hurts choice prediction. Bottom left: in-sample against out-of-sample RT fit, showing both that the run A ranking does not generalise and how much every model loses between sessions. Bottom middle: recovery RMSE ratio against the choice-only fit, where below the dashed line means RT improves recovery. Bottom right: test–retest reliability of $\log h_{gain}$ per RT model against the choice-only baseline (dashed).

TABLES

Table 1: Priors on the fitted (transformed) scale, and their rationale.

Parameter	Prior	Rationale
<code>log_h_gain</code> , <code>log_h_loss</code>	$\mathcal{N}(0, 2^2)$	Centred at $h = 1$, with 95% mass over $h \in [0.02, 55]$, far wider than the model-free estimates (3.24, 0.35), so the data dominate.
<code>log_beta</code>	$\mathcal{N}(1, 1.5^2)$	Covers $\beta \in [0.14, 52]$; excludes only numerically degenerate values.
<code>beta0</code>	$\mathcal{N}(0, 2^2)$	No stance on the direction of bias.
<code>a</code>	$\mathcal{N}(0.5, 1^2)$	Centred near $\log(\text{median RT} - \tau) \approx 0.6$.
<code>b</code>	$\mathcal{N}(0, 1^2)$; $\mathcal{N}(0, 3^2)$ for M3	Centred at zero and symmetric, so a positive difficulty effect is not assumed. Widened for M3 because ΔV^2 is on a smaller scale, which keeps regularisation comparable in effect rather than in raw coefficient.
<code>log_sigma</code>	$\mathcal{N}(-0.5, 0.5^2)$	Matches the observed spread of $\log$ RT.
<code>tau_raw</code>	$\mathcal{N}(0, 1.5^2)$	Nearly flat over the admissible interval $(0, \min RT)$.

Table 2: Parameter recovery over 200 simulated participants per model. r and RMSE are between true and recovered values from the full choice + RT fit; the RMSE ratio is (joint fit)/(choice-only fit), so values below 1 mean RT improves recovery. b 's RMSE is not comparable across models because the predictors are on different scales.

Parameter	M1 (linear)			M2 (uncertainty)			M3 (quadratic)		
	r	RMSE	ratio	r	RMSE	ratio	r	RMSE	ratio
$\log h_{gain}$	0.925	0.569	0.862	0.900	0.649	0.983	0.977	0.319	0.471
$\log h_{loss}$	0.958	0.486	0.719	0.936	0.598	0.883	0.978	0.357	0.490
$\log \beta$	0.955	0.266	0.964	0.958	0.253	0.917	0.956	0.263	0.923
β_0	0.954	0.371	0.869	0.952	0.382	0.895	0.963	0.326	0.760
a	0.884	0.107	–	0.825	0.145	–	0.928	0.086	–
b	0.897	0.184	–	0.860	0.210	–	0.978	0.238	–
$\log \sigma$	0.917	0.097	–	0.914	0.098	–	0.935	0.088	–
τ	0.187	0.110	–	0.207	0.103	–	0.287	0.089	–

Table 3: Test–retest reliability (Pearson r between run A and run B estimates, $n = 49$). Δ is the difference from the choice-only baseline, and bracketed values are bootstrap 95% CIs on that difference (2000 resamples, paired across models). Dashes mark parameters that do not exist in the baseline.

Parameter	baseline	M1	M2	M3	Δ M3 vs baseline
$\log h_{gain}$	0.444	0.212	0.187	0.447	+0.003 [−0.21, +0.23]
$\log h_{loss}$	0.124	0.118	0.099	0.304	+0.180 [−0.04, +0.40]
$\log \beta$	0.575	0.560	0.504	0.532	−0.043
β_0	0.361	0.092	0.164	0.276	−0.085
a	–	0.682	0.617	0.727	–
b	–	0.089	0.200	0.100	–
$\log \sigma$	–	0.666	0.671	0.669	–
τ	–	0.414	0.391	0.407	–

Table 4: Out-of-sample performance on run B, participant means. Run A parameters are applied unchanged. RT models are compared with each other on LL_{RT} only; RT-informed fits are compared with the baseline on LL_{choice} and balanced accuracy only. All log-likelihoods are per trial, and chance for choice is -0.693 .

Model	Condition	LL_{choice}	bal. acc.	AUC	LL_{RT}	RMSE logRT
baseline	all	-0.630	0.699	0.756	–	–
	reward	-0.637	0.689	0.744	–	–
	loss	-0.623	0.697	0.767	–	–
M1	all	-0.640	0.692	0.752	-1.548	0.775
	reward	-0.650	0.690	0.746	-1.616	0.794
	loss	-0.629	0.690	0.770	-1.481	0.741
M2	all	-0.643	0.690	0.750	-1.524	0.761
	reward	-0.651	0.683	0.739	-1.546	0.769
	loss	-0.634	0.692	0.774	-1.502	0.740
M3	all	-0.640	0.689	0.754	-1.544	0.781
	reward	-0.646	0.689	0.749	-1.596	0.797
	loss	-0.633	0.687	0.764	-1.492	0.748

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